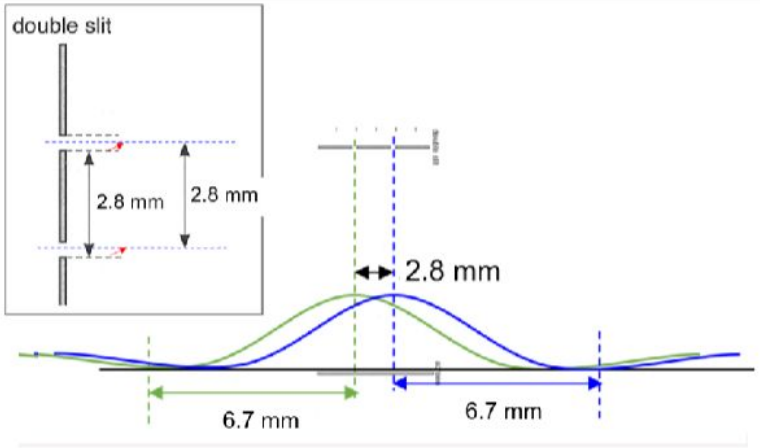
	TOTAL FOR QUESTION 4	**9**
5(a)	For a single-slit diffraction pattern, $\sin\theta = \frac{\lambda}{b} \Rightarrow \theta = \sin^{-1}\left(\frac{\lambda}{b}\right) = \sin^{-1}\left(\frac{670\times10^{-9}}{0.12\times10^{-3}}\right) = 0.00558336 \text{ rad (6.s.f)}$ Let the distance between the central maxima to first minima for the diffraction pattern be *y*. Hence, $\tan\theta = \frac{y}{D} \Rightarrow y = D\ (\tan\theta)$ $= 1.20\ (\tan(0.00558336))$ $= 0.00670010 \text{ m (6.s.f)}$ Thus, the width of the central fringe is: $2y = 2\ (0.00670010)$ $= 0.0134 \text{ m (3.s.f)}$ $= 13.4 \text{ mm (shown)}$ **Note:** Deduct 1 mark if students - used small angle approximation without justification - write the angle in degrees without indicating the degree symbol	**C1** **C1** **M1** **A0**
5(b)(i)	The distance from the central bright maximum to first minimum is: 13.4 / 2 = 6.7 mm. However, the central bright maxima are 2.8 mm away from each other. Hence, we cannot distinguish between the two diffraction patterns from each single slit / they cannot be resolved.	**M1** **A1**

Qns	Answer	Marks
		
5(b)(ii)	$x = \frac{D\lambda}{a} = \frac{(1.2)(670\times10^{-9})}{(2.8\times10^{-3})} = 0.000287143 \text{ m (6.s.f)}$ An estimation of the number of fringes, n is $n = \frac{0.0134002}{0.000287143} = 46.66733 = 46$ (round down)	C1 A1
	TOTAL FOR QUESTION 5	**7**
6(a)	Faraday's law of electromagnetism states that the (induced) e.m.f. is directly proportional to	B1